

Elimination of Left Recursion

A grammar is left recursive if it has a nonterminal A such that there is a derivation $A \Rightarrow^+ A\alpha$ for some string α .

Top down parsing method cannot handle left-recursive grammar. So, we eliminate left recursion. The immediate left recursion is in this form $A \rightarrow A\alpha$.

Suppose a grammar is $A \rightarrow A\alpha | \beta$.

Here left recursive pair of production can be replaced by the following non-left recursive productions.

$$A \rightarrow \beta A'$$

$$A' \rightarrow \alpha A'$$

$$A' \rightarrow \epsilon$$

change the following grammar into non-left recursive grammar

$$E \rightarrow E + T | T$$

$$T \rightarrow T * F | F$$

$$F \rightarrow (E) | id$$

$$E \rightarrow TE' \quad E' \rightarrow +TE' \quad E' \rightarrow \epsilon$$

$$T \rightarrow FT' \quad T' \rightarrow *FT' \quad T' \rightarrow \epsilon$$

$$F \rightarrow (E) \quad F \rightarrow id$$

Immediate left recursion can be eliminated by the following technique, which works for any number of A -productions.

$$A \rightarrow A\alpha_1 | A\alpha_2 | A\alpha_3 | \dots | A\alpha_m | \beta_1 | \beta_2 | \dots | \beta_n$$

$$A \rightarrow \beta_1 A' | \beta_2 A' | \beta_3 A' | \dots | \beta_n A'$$

$$A' \rightarrow \alpha_1 A' | \alpha_2 A' | \alpha_3 A' | \dots | \alpha_m A'$$

$$A' \rightarrow \epsilon$$

$$S \rightarrow Aa|b$$

$$A \rightarrow Ac|Sd|\epsilon$$

- 1) arrange the nonterminals in some order
 $A_1, A_2, \dots, A_n$
- 2) for (each i from 1 to n) {
- 3 for (each j from 1 to $i-1$) {
- 4 replace each production of the form $A_i \rightarrow A_j \gamma$
 by the production $A_i \rightarrow \delta_1 \gamma | \delta_2 \gamma | \dots | \delta_k \gamma$,
 where $A_j \rightarrow \delta_1 | \delta_2 | \dots | \delta_k$ are all current
 A_j -productions
- 5 }
- 6 eliminate the immediate left recursion among
 the A_i -productions
- 7 }

$$S \rightarrow Aa|b$$

$$A \rightarrow bdA' | A'$$

$$A' \rightarrow cA' | adA' | \epsilon$$

$$S \rightarrow Aa|b$$

$$A \rightarrow Ac | Aad | bd | e$$

$$S \rightarrow Aa|b$$

$$A \rightarrow bdA' | eA'$$

$$A' \rightarrow cA' | adA' | \epsilon$$

Left Factoring

In general, if $A \rightarrow \alpha\beta_1 \mid \alpha\beta_2$ are two A -productions and the input string begins with a non empty string derived from α , we do not know whether to expand A to $\alpha\beta_1$ or $\alpha\beta_2$.

$$A \rightarrow \alpha A'$$

$$A' \rightarrow \beta_1 \mid \beta_2$$

$$A \rightarrow \alpha\beta_1 \mid \alpha\beta_2 \mid \dots \mid \alpha\beta_n \mid \gamma$$

$$A \rightarrow \alpha A' \mid \gamma$$

$$A' \rightarrow \beta_1 \mid \beta_2 \mid \dots \mid \beta_n$$

$$S \rightarrow iEtS \mid iEtSes \mid a$$

$$E \rightarrow b$$

$$S \rightarrow iEtSS' \mid a$$

$$S' \rightarrow eS \mid \epsilon$$

$$E \rightarrow b$$